

Srikrishnaa Vadivel | Electrical Engineer | Photonics | Embedded Systems | Scientific Computing

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Profile

Electrical engineer with Cornell ECE training, photonics communication, embedded systems delivery, FPGA/MCU experience, and current Yale PhD work in computational materials and high-performance scientific software.

Experience

Yale University

PhD Researcher / Embedded Systems TA

2021–present

- Research first-principles self-trapped excitons using HPC systems and scientific software stacks.
- Teach and support embedded systems concepts spanning firmware, instrumentation, and debugging.

Probe Technologies / Weatherford

Software and Embedded Engineer

2019–2021

- Developed embedded sensor logging firmware and Lattice FPGA logic for geological well tools.
- Built CUDA/OpenGL acoustic waveform processing and visualization tooling.
- Contributed ASP and Microsoft database software for field data workflows.

FindLight

Photonics Blogger

2018–2019

- Wrote clear technical content on photonics products, optics, and applications.

Projects

Digital Design: Verilog MIPS CPU and matrix-multiply accelerator with Icarus test benches.

Photonics: Meep ring resonator FDTD model for integrated photonics simulation.

Numerics: DOLFINx PDE notes and solver scaffold for finite-element methods.

Robotics: Embedded robotic hand project published in Circuit Cellar.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BA, Electrical and Computer Engineering

2013–2017

Skills

Core: Circuits, embedded C, Verilog, FPGA, MCUs, signal processing

Photonics: Photonics, optics communication

Software: Python, C++, Linux, OpenGL, CUDA

Scientific: HPC, numerical methods